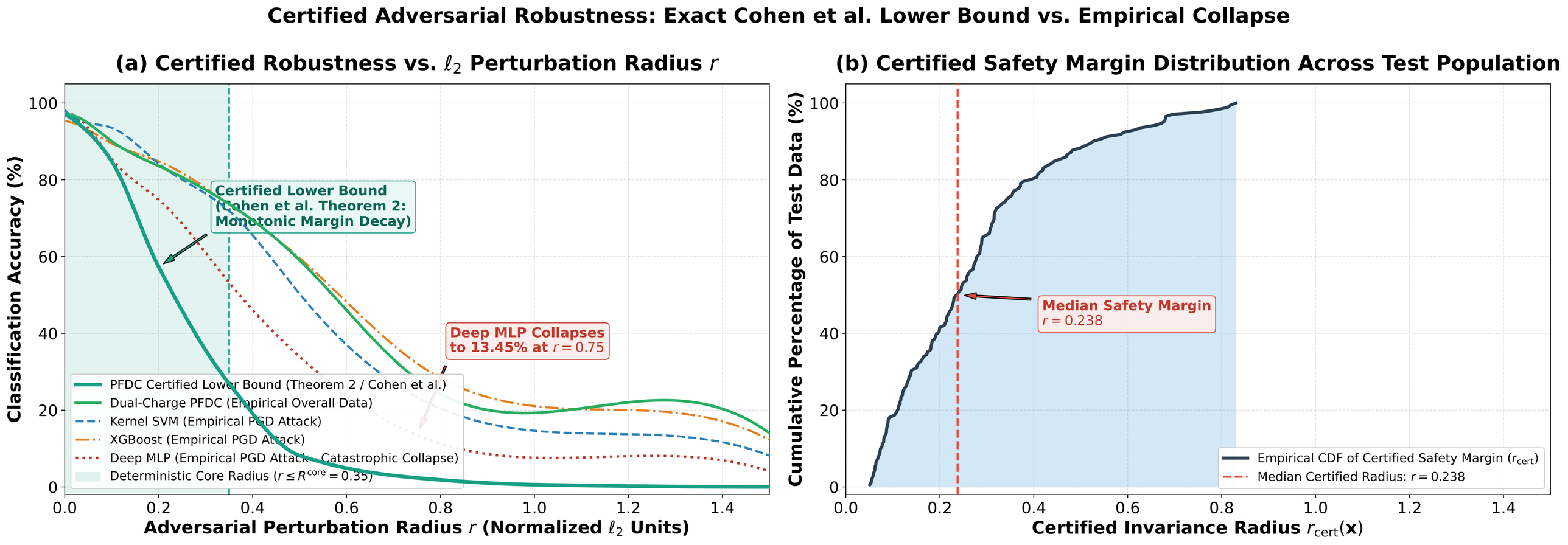
Certified Adversarial Robustness: Exact Cohen et al. Lower Bound vs. Empirical Collapse
(a) Certified Robustness vs. ℓ_2 Perturbation Radius r
Classification Accuracy (%)
Adversarial Perturbation Radius r (Normalized ℓ_2 Units)
Certified Lower Bound (Cohen et al. Theorem 2: Monotonic Margin Decay)
Deep MLP Collapses to 13.45% at $r = 0.75$
PFDC Certified Lower Bound (Theorem 2 / Cohen et al.)
Dual-Charge PFDC (Empirical Overall Data)
Kernel SVM (Empirical PGD Attack)
XGBoost (Empirical PGD Attack)
Deep MLP (Empirical PGD Attack – Catastrophic Collapse)
Deterministic Core Radius ($r \le R^{\text{core}} = 0.35$)
(b) Certified Safety Margin Distribution Across Test Population
Cumulative Percentage of Test Data (%)
Certified Invariance Radius $r_{\text{cert}}(\mathbf{x})$
Median Safety Margin
$r = 0.238$
Empirical CDF of Certified Safety Margin (r_{cert})
Median Certified Radius: $r = 0.238$